

Technical Details

Unreal Engine uses variable frame rates. While it is good to improve the scalability of the hardware, it creates challenges for physics engine that works best with a small and fixed number of steps. So, what Sub-stepping does is that it takes the total number of frames needed to execute a particular full step and divides them into numerous sub-steps.

The number of sub-steps will depend on the combination of Max Sub-steps Property as well as Max Sub-Step Delta Time defined for the particular set of subsets.

Sub-Stepping in Unreal Engine is completely hidden from the main user of the game, which means one needs to interpolate or maintain some calls to the physical engine.

Uses of Physical Sub-Stepping

Sub-Stepping drastically improves the quality of complex physical assets that are used. However, you as a game developer should always be mindful of the options, especially when they come at the cost of your CPU speed and accordingly determines what your game needs more under the given circumstances.

Materials

Material is the property that applies to a mesh to control the visual appearance of a scene. At a high level, it is easy to think of an object as “paint”.

But this is also a bit misleading because a material literally defines the type of surface that your object is going to create. You can define its colour, how bright it is, what you can see through the object, and more.

In other words, when a light touches the surface from the scene, a material is used to calculate how light relates to that surface.

Material Editor

Material Editor is a node-based graph interface that allows you to create shades that apply to your geometry with static and skilnet meshes or other systems such as Cascade to create interesting content.

Content Editor Opening

The Content Editor opens by double-clicking on any physical property or the right-click context menu of the Content property in the Content Browser. When you do any of these two operations, that particular content is opened in the Content Editor for editing.